Decision Memo: Enhanced Service Quality Checks Pilot

Nov 7, 2026

Recommendation: EXTEND. Do not roll out chain-wide yet. Results at the Noida multi-brand service chain (18 dealerships) favour the initiative, but the test cannot yet prove it caused them. Extend the evaluation by about two weeks for a dealership-level re-analysis, and roll out to the 9 control dealerships in randomised waves, then re-decide go or no-go against the criteria below.

Why. Treated dealerships (D01–D09) and control dealerships (D10–D18) never overlap, so the real comparison is 9 vs 9 dealerships, not 45,000 vehicles. Row-level confidence intervals are too narrow, and any pre-existing difference between the two groups of dealerships is indistinguishable from the treatment effect.

Metric	Control	Treatment	Difference
Warranty claim rate (primary)	9.50%	8.04%	–1.46 pp
30-day repeat visits	13.99%	11.06%	–2.93 pp
Post-period revenue per vehicle	₹7,881	₹8,255	+₹374

Intervals are omitted because the notebook's row-level intervals are not valid for this design.

What could make the causal estimate wrong

- **Dealership confounding:** location, brand mix, customer base or workshop quality differ between the two groups.
- **Non-random assignment:** treatment went to the first nine dealership IDs and control to the last nine.
- **Concurrent shock:** claims rose in 2026 among non-treated rows, possibly linked to the supplier part change (PART_OLD to PART_NEW).
- **Historical DiD unreliable:** mis-mapped group indicator and untested parallel trends, so it is not relied on.
- **Data quality:** claim-to-job linkage unverified, 18% of failure codes missing, part-code crosswalk matched nothing.
- **Immature or missing outcomes:** recent periods may lack full follow-up; complaints, duration and refunds are unmeasured.

Go only if all hold:

1. A dealership-level, baseline-adjusted 95% interval for the claim-rate change excludes zero and meets the minimum effect [placeholder: 1 pp].
2. Guardrails stay within limits set in advance [thresholds to be agreed].
3. Claim linkage is documented and part-supplier mix is balanced across arms.
4. The full 60-day follow-up has elapsed.

No-go if the adjusted estimate is near zero or adverse, or a guardrail is breached.

Next two weeks: confirm how dealerships were assigned and the start date; estimate the intracluster correlation and run an exact permutation test (48,620 possible assignments); rebuild the DiD; fix data linkage; set guardrail thresholds; design the randomised waves.